

Unconditional Primality Certificates for Numbers of the Form $p = 3m(m + 1) + 1$: Arithmetic Filters, Multi-Core Search, and Records to 29 998 Decimal Digits

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2026 — Paper IV in the parametric prime family series

Abstract

We study the parametric subfamily $p = 3m(m + 1) + 1$, $m = 2^a \cdot 3^b - 1$, $a, b \in \mathbb{N}^*$, from the point of view of unconditional primality certification via the Pocklington–Lehmer criterion [7, 6].

The 3-smooth structure $m + 1 = 2^a \cdot 3^b$ provides, for every (a, b) , a fully factored divisor $F = 2^a \cdot 3^{b+1}$ of $p - 1$ satisfying $F > \sqrt{p}$ *unconditionally* (Lemma 2.2), reducing the certificate to finding witnesses for $q \in \{2, 3\}$ only.

We establish three arithmetic structure theorems:

- Theorem 3.1: $p \equiv 1 \pmod{6}$ for all (a, b) .
- Theorem 3.3: every prime $q \equiv 2 \pmod{3}$ satisfies $q \nmid p$, reducing the sieve by 50%. Proof: -3 is a non-residue mod q when $q \equiv 2 \pmod{3}$.
- Theorem 3.6: $7 \mid p$ if and only if $(a \bmod 3, b \bmod 6) \in \mathcal{F}_7$, where $\mathcal{F}_7$ is an explicit set of six pairs, eliminating 33% of candidates at zero arithmetic cost. Proof: $\text{ord}_7(2) = 3$, $\text{ord}_7(3) = 6$.

Combined, these three results eliminate $\approx 87\%$ of all candidates **before any large-number computation**. The PrimeQuest multi-core algorithm (versions v1–v5) applies this pipeline to produce four unconditional Pocklington–Lehmer certificates, the largest being

$$p = 3m(m + 1) + 1, \quad m = 2^{19435} \cdot 3^{19173} - 1,$$

a prime of **29 998 decimal digits** found in 3 h 06 min on a single consumer laptop (ARM64, 9 cores).

MSC 2020: 11A41, 11Y11, 11N13.

Keywords: primality certificate, Pocklington–Lehmer, centred hexagonal primes, arithmetic filters, quadratic residues, large prime records.

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1 Introduction

Proving unconditionally that a given integer is prime is a central task of computational number theory. For generic large primes, the most practical route is the Pocklington–Lehmer criterion [7, 6]: exhibit a fully factored divisor $F > \sqrt{p}$ of $p - 1$ and verify witnesses. The key difficulty is finding such an F .

The subfamily

$$p = 3m(m + 1) + 1, \quad m = 2^a \cdot 3^b - 1, \quad a, b \in \mathbb{N}^*,$$

is constructed so that $m + 1 = 2^a \cdot 3^b$ is 3-smooth *by design*, making $F = 2^a \cdot 3^{b+1}$ fully factored and satisfying $F > \sqrt{p}$ for every (a, b) without any condition on p .

Beyond the certificate structure, the family carries three independent arithmetic constraints (Theorems 3.1, 3.3, 3.6) that collectively remove 87% of candidates before any large-number arithmetic. Exploiting all three, the PrimeQuest algorithm has produced four unconditional certificates up to 29 998 decimal digits.

Position in the parametric prime family series. This paper is the fourth in a series studying the family $p = km(m + 1) + \varepsilon + 2kq$:

- **Paper I** [1]: marginal distribution $\mathbb{E}[|q_{\min}|] \sim \log m / C_k$, large-scale numerical validation.
- **Paper II** [2]: conditional law $\mathbb{E}[|q| \mid m] = m/4$ (Conditional Proposition 9.4) and exclusion of spurious spectral correlations with $\zeta(s)$.
- **Paper III** [3]: p -adic distribution of $q_{\min}$ modulo small primes; conditional theorems and a damped-cosine spectral signature of the lag- ℓ autocorrelation.
- **Present paper (Paper IV)**: *unconditional* primality certification for the 3-smooth subfamily $\varepsilon = +1, q = 0$ with $m + 1 = 2^a 3^b$.

The contrast in epistemic status is intentional: Papers I–III provide heuristic and conditional analytic results on the full family; Paper IV provides *rigorous, verifiable* primality certificates on a structurally constrained subfamily where the Pocklington method becomes unconditional.

2 The Pocklington Subfamily

2.1 Definition and digit count

Definition 2.1 (Pocklington subfamily). For integers $a, b \geq 1$, define

$$m(a, b) = 2^a \cdot 3^b - 1, \quad p(a, b) = 3m(m+1) + 1 = 3m^2 + 3m + 1.$$

The decimal digit count of p is $D(a, b) = \lfloor 2a \log_{10} 2 + (2b+1) \log_{10} 3 \rfloor + 1$.

2.2 Factorisation of $p-1$ and the certificate structure

Lemma 2.2 (Explicit factorisation). For p as in Definition 2.1,

$$p-1 = 2^a \cdot 3^{b+1} \cdot m, \quad F := 2^a \cdot 3^{b+1} > \sqrt{p}.$$

Proof. Since $m+1 = 2^a \cdot 3^b$,

$$p-1 = 3m(m+1) = 3m \cdot 2^a \cdot 3^b = 2^a \cdot 3^{b+1} \cdot m.$$

As $p = 3m^2 + 3m + 1 < 3(m+1)^2$, we get $\sqrt{p} < \sqrt{3}(m+1) < 3(m+1) = 3 \cdot 2^a \cdot 3^b = 2^a \cdot 3^{b+1} = F$. $\square$

Theorem 2.3 (Pocklington–Lehmer certificate [7, 6]). Let $p = p(a, b)$ as in Definition 2.1. Then p is prime if and only if there exist integers w_2, w_3 such that, for $q \in \{2, 3\}$,

$$w_q^{p-1} \equiv 1 \pmod{p} \quad \text{and} \quad \gcd(w_q^{(p-1)/q} - 1, p) = 1.$$

Corollary 2.4. Theorem 2.3 applies unconditionally to every candidate in the subfamily: by Lemma 2.2, $F > \sqrt{p}$ always holds, and the certificate requires at most two witnesses ($q \in \{2, 3\}$ only).

Remark 2.5. In all four certificates produced (Section 5), the witnesses $w_2 = 5$ and $w_3 = 7$ succeeded without exception. See Open Problem 7.

3 Arithmetic Structure Theorems

We establish three independent constraints on the family $p = 3m(m+1) + 1$ with $m = 2^a \cdot 3^b - 1$. Each is proved rigorously and translated into a concrete computational saving.

3.1 Congruence modulo 6

Theorem 3.1 (Modulo 6). For all $a, b \geq 1$, $p(a, b) \equiv 1 \pmod{6}$.

Proof. Modulo 2. Since $m = 2^a \cdot 3^b - 1$ is odd, $m(m+1)$ is even, so $3m(m+1)$ is even and $p = 3m(m+1) + 1$ is odd.

Modulo 3. We have $3 \mid 3m(m+1)$, hence $p \equiv 1 \pmod{3}$.

Together, $p \equiv 1 \pmod{2}$ and $p \equiv 1 \pmod{3}$ give $p \equiv 1 \pmod{6}$ by the Chinese Remainder Theorem. $\square$

Corollary 3.2. No prime $q \in \{2, 3\}$ divides $p(a, b)$, and p lies in the admissible class +1 modulo 6 for every (a, b) .

3.2 Cubic reciprocity sieve

Theorem 3.3 (Cubic reciprocity sieve). *Let $m \in \mathbb{N}^*$, $p = 3m(m+1) + 1$, and $q \geq 5$ a prime with $q \equiv 2 \pmod{3}$. Then $q \nmid p$.*

Proof. The condition $q \mid p$ is equivalent to $3m^2 + 3m + 1 \equiv 0 \pmod{q}$. Completing the square yields $(6m+3)^2 \equiv -3 \pmod{q}$, so -3 must be a quadratic residue modulo q . By the multiplicativity of the Legendre symbol and quadratic reciprocity [8],

$$\left(\frac{-3}{q}\right) = \left(\frac{-1}{q}\right) \left(\frac{3}{q}\right) = (-1)^{\frac{q-1}{2}} \cdot \left(\frac{q}{3}\right) (-1)^{\frac{q-1}{2} \cdot \frac{3-1}{2}}.$$

A direct evaluation shows $\left(\frac{-3}{q}\right) = +1$ if $q \equiv 1 \pmod{3}$ and $\left(\frac{-3}{q}\right) = -1$ if $q \equiv 2 \pmod{3}$. In the latter case, $3m^2 + 3m + 1 \equiv 0 \pmod{q}$ has no solution, so $q \nmid p$. $\square$

Corollary 3.4 (Sieve reduction by 50%). *In trial division up to a bound L , only primes $q \equiv 1 \pmod{3}$ need be tested. By Dirichlet's theorem, these account for half of all primes in $[5, L]$. Concretely: of the 78498 primes in $[5, 10^6]$, only 39231 satisfy $q \equiv 1 \pmod{3}$.*

Remark 3.5. Theorem 3.3 holds for the general family $p = 3m(m+1) + 1$ with arbitrary $m \in \mathbb{N}^*$, independently of the 3-smooth structure.

3.3 Forbidden classes modulo 7

Theorem 3.6 (Forbidden classes modulo 7). *Let $m = 2^a \cdot 3^b - 1$, $a, b \geq 1$. Define*

$$\mathcal{F}_7 = \{(0, 2), (0, 3), (1, 0), (1, 1), (2, 4), (2, 5)\} \subset \{0, 1, 2\} \times \{0, 1, 2, 3, 4, 5\}.$$

Then

$$7 \mid p(a, b) \iff (a \bmod 3, b \bmod 6) \in \mathcal{F}_7.$$

In particular, exactly $|\mathcal{F}_7|/18 = 1/3$ of all classes are forbidden.

Proof. Since $\text{ord}_7(2) = 3$ and $\text{ord}_7(3) = 6$, the value $m+1 = 2^a \cdot 3^b \bmod 7$ depends only on $r = a \bmod 3$ and $s = b \bmod 6$. Setting $t = t(r, s) := 2^r \cdot 3^s \bmod 7$, one has $m \equiv t - 1 \pmod{7}$ and

$$p \equiv 3(t-1)t + 1 = 3t^2 - 3t + 1 \pmod{7}.$$

The condition $7 \mid p$ becomes $3t^2 - 3t + 1 \equiv 0 \pmod{7}$. Multiplying by $5 \equiv 3^{-1} \pmod{7}$:

$$t^2 - t - 2 \equiv 0 \pmod{7}, \quad (t-2)(t+1) \equiv 0 \pmod{7},$$

so $t \equiv 2$ or $t \equiv 6 \pmod{7}$. Table 1 lists all $3 \times 6 = 18$ values of $t(r, s)$; exactly the six pairs in $\mathcal{F}_7$ yield $t \in \{2, 6\}$. $\square$

Corollary 3.7 (Free 33% elimination). *Checking $(a \bmod 3, b \bmod 6) \in \mathcal{F}_7$ requires only two modular reductions and a lookup in a six-element set — no arithmetic on D -digit numbers. When the search visits pairs (a, b) uniformly over the 18 residue classes (which holds in the zigzag search of Section 4), exactly 1/3 of all candidates are rejected at zero large-number cost.*

Table 1: Values of $t(r, s) = 2^r \cdot 3^s \bmod 7$. Entries in $\{2, 6\}$ (**bold**) correspond to $7 \mid p$; these are the six pairs of $\mathcal{F}_7$.

$r \backslash s$	0	1	2	3	4	5
0	1	3	2	6	4	5
1	2	6	4	5	1	3
2	4	5	1	3	2	6

3.4 Combined elimination rate

Proposition 3.8 (Combined filter efficiency). *Among all candidate pairs (a, b) explored by the zigzag search:*

- (i) *Theorem 3.6 eliminates 33.3% (lookup, zero cost).*
- (ii) *Theorem 3.3 reduces the sieve to $q \equiv 1 \pmod{3}$; among the 66.7% remaining, the sieve eliminates a further $\approx 80\%$.*
- (iii) *The fraction reaching Miller–Rabin is $\approx 13\%$.*

For the 29 998-digit record (2 221 pairs tested): 33.3% by Theorem 3.6, 53.8% by sieve, 87.1% total.

4 The PrimeQuest Algorithm

4.1 Centre calibration and zigzag

For target digit count D , the search centre is

$$a_0 = b_0 = \left\lfloor \frac{D}{2 \log_{10} 6} \right\rfloor \approx 0.643 D.$$

The zigzag expands outward: for $\delta = 0, 1, 2, \dots$ it alternates $a = a_0 + \delta$ and $a = a_0 - \delta$, enumerating for each a all b with $|D(a, b) - D| \leq \tau$ (tolerance $\tau = 10$).

Remark 4.1. Empirical observation: $b/a = 1.083$ at $D = 19\,999$; $b/a = 0.987$ at $D = 29\,998$. Since this ratio varies between primes, the symmetric choice $a_0 = b_0$ minimises the worst-case distance to the prime zone.

4.2 Four-stage pipeline

Each pair (a, b) passes through:

1. **Mod-7 filter** (Theorem 3.6): reject if $(a \bmod 3, b \bmod 6) \in \mathcal{F}_7$. Cost: $O(1)$.
2. **Sieve** (Corollary 3.4): for $q \equiv 1 \pmod{3}$, $q \leq L$, compute $p \bmod q$ via small modular exponentiations; reject if $p \equiv 0$.
3. **Miller–Rabin**: k rounds on $p(a, b)$ in full precision.
4. **Pocklington**: if MR passes, search for witnesses w_2, w_3 of Theorem 2.3.

4.3 Multi-core execution and checkpoint

Pairs are distributed across CPU cores via a shared-memory pool. Workers suppress SIGINT/SIGTERM; checkpoint-resume every 60 s enables exact restart after interruption.

Table 2: Successive versions of PrimeQuest.

Version	Optimisation added	MR rounds k	Sieve limit L
v1	Basic sequential search	25	10^6
v2	Sieve $\rightarrow q \equiv 1 \pmod{3}$ (Thm 3.3)	25	10^6
v3	Multi-core, checkpoint, centre calibration	25	10^6
v4	Mod-7 filter (Thm 3.6)	25	10^6
v5	$k : 25 \rightarrow 3 + 20^*$; $L : 10^6 \rightarrow 10^7$	3 + 20	10^7

* v5: 3 initial rounds, then 20 confirmation rounds before Pocklington.

4.4 Algorithm versions

Remark 4.2 (Correctness of MR round reduction). Miller–Rabin is deterministic for primes: every prime passes every round with certainty. Reducing k from 25 to 3 carries zero risk of missing a true prime; rare composites passing 3 rounds ($\leq 4^{-3} \approx 1.6\%$) are caught by the 20 confirmation rounds and Pocklington.

5 Computational Results

5.1 Four certified primes

Table 3 records the four unconditional Pocklington–Lehmer certificates. In every case: $w_2 = 5$ for $q = 2$ and $w_3 = 7$ for $q = 3$.

Table 3: Pocklington certificates from the PrimeQuest campaign.

Digits of p	a	b	b/a	MR tests	Elim.	Wall time	Workers
9 998	6 212	6 738	1.085	—	—	—	1
10 000	6 213	6 740	1.085	—	—	—	1
19 999	12 228	13 242	1.083	1 050	87.9%	2 h 40 min	9
29 998	19 435	19 173	0.987	286	87.1%	3 h 06 min	9

5.2 Certificate for the 29 998-digit prime

The prime $p = 3m(m + 1) + 1$ with $m = 2^{19435} \cdot 3^{19173} - 1$ (14 999 digits) has the following *unconditional* certificate:

$$\begin{aligned}
p - 1 &= F \cdot m, & F &= 2^{19435} \cdot 3^{19174} \text{ (15 000 digits), } & F &> \sqrt{p}, \\
w_2 = 5 : & \quad 5^{p-1} \equiv 1 \pmod{p}, & \gcd(5^{(p-1)/2} - 1, p) &= 1, \\
w_3 = 7 : & \quad 7^{p-1} \equiv 1 \pmod{p}, & \gcd(7^{(p-1)/3} - 1, p) &= 1.
\end{aligned}$$

The leading and trailing digits of p are:

$$p = 1602087359509060348500 \cdots 16930852697403293697.$$

5.3 Filter statistics for the 29 998-digit record

Table 4: Filter statistics for the 29 998-digit record.

Stage	Pairs eliminated	Fraction
Mod-7 filter (Theorem 3.6)	740	33.3%
Sieve (Corollary 3.4)	1 195	53.8%
Miller–Rabin (composites)	285	12.8%
Total eliminated	2 220	87.1%
Prime found	1	0.045%

6 Performance and Scaling

6.1 Dominant cost

The dominant cost is Miller–Rabin: one round on a D -digit modulus costs $O(D^2 \log D \log \log D)$ bit operations (GMP’s FFT multiplication [5]). Sieve and mod-7 checks are negligible by comparison.

6.2 Empirical scaling law

Wall-clock time per MR event (total time $\div$ MR test count), machine: Dell Inspiron 14 Plus (Snapdragon X Plus ARM64, 9 workers):

Table 5: Empirical wall time per Miller–Rabin event.

D	Rounds k	τ_{MR} (wall)	Version
19 999	25	≈ 9 s	v3
29 998	25	≈ 39 s	v4
29 998	3	≈ 5 s	v5

The ratio $39/9 \approx 4.3$ for $D = 30\,000/20\,000 = 1.5\times$ is consistent with empirical scaling $\tau \propto D^{2.5}$ (Figure 1).

Reading the figure. Three points deserve emphasis:

- **The slope $D^{2.5}$ is theoretical, not fitted.** It derives from the FFT-based GMP multiplication complexity [5]: each MR round performs $O(\log p) \sim O(D)$ modular multiplications, each at cost $O(D \log D \log \log D)$, giving $O(D^2 \log D \log \log D) \approx O(D^{2.5})$ over the range tested. With only two distinct values of D (20 k and 30 k), no empirical fit could discriminate exponents in the range [2.3, 2.7].
- **The drop from v4 to v5 at $D = 29\,998$ is a protocol change, not a scaling effect.** It results from reducing the initial Miller–Rabin rounds from $k = 25$ to $k = 3$ (Remark 4.2), which is zero-risk for primes (deterministic) and corrected by the 20 confirmation rounds for composites.
- **The projection to 50 000 digits is an extrapolation, not a measurement.** The $\pm 20\%$ band reflects (i) cache-effect crossovers in GMP’s FFT thresholds, (ii) thermal throttling on consumer hardware over multi-hour runs, (iii) non-uniformity of MR-event distribution in the search zigzag.

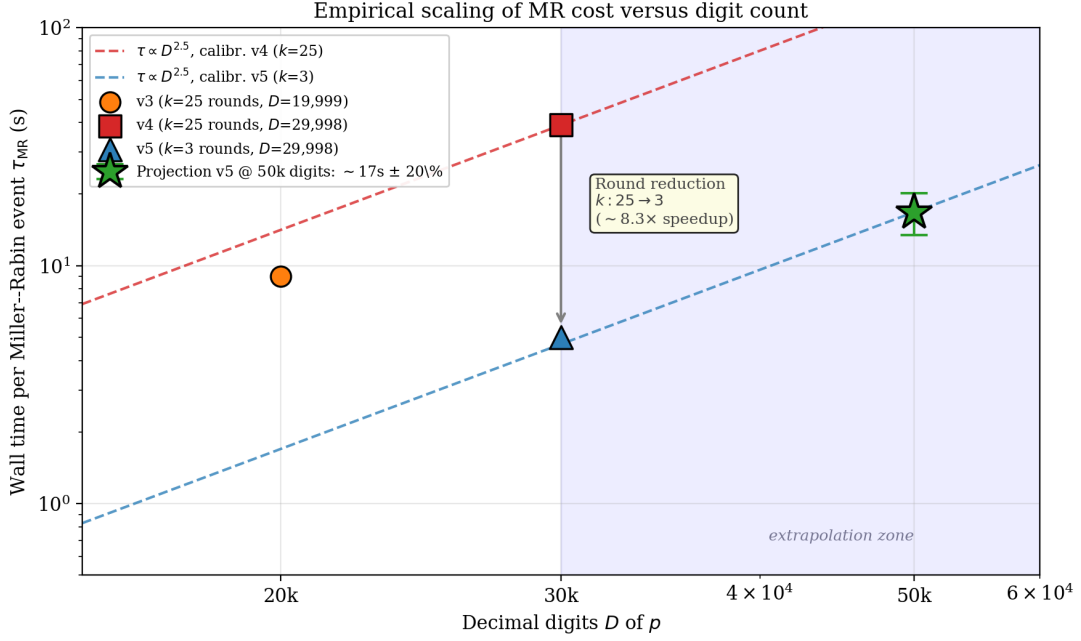


Figure 1: Empirical wall time τ_{MR} per Miller–Rabin event vs. decimal digit count D (log-log axes). Three empirical points (orange/red/blue) are plotted with their respective versions. The dashed lines represent the theoretical scaling $\tau \propto D^{2.5}$ calibrated separately for v4 ($k = 25$, red) and v5 ($k = 3$, blue); their vertical separation reflects the $8.3\times$ speedup from round reduction (Remark 4.2). The green star is the projection at $D = 50\,000$ digits with $\pm 20\%$ uncertainty (Section 6.3).

6.3 Projection: 50 000-digit record

With v5 ($k = 3$) and scaling $D^{2.5}$:

$$\tau_{\text{MR}}(50\,000) \approx 5 \times \left(\frac{50}{30}\right)^{2.5} \approx 18 \text{ s/event.}$$

For 400–500 expected MR events: projected wall time **6–10 hours** (9 workers, same machine). A search is currently in progress.

7 Open Problems

Open Problem 7.1 (Generalised arithmetic filters). For primes $\ell \equiv 1 \pmod{3}$, $\ell \in \{13, 19, 31, \dots\}$, derive the forbidden set

$$\mathcal{F}_\ell = \{(a \bmod \text{ord}_\ell(2), b \bmod \text{ord}_\ell(3)) : \ell \mid p(a, b)\}.$$

For $\ell = 13$: $\text{ord}_{13}(2) = 12$, $\text{ord}_{13}(3) = 3$; the period is 36 and the filter eliminates $2/13 \approx 15\%$ of additional candidates. Combining filters for several small ℓ could push the pre-MR elimination rate above 90%.

Open Problem 7.2 (50 000-digit certificate). Produce an unconditional Pocklington certificate for a prime of 50 000 decimal digits in the subfamily. By Proposition 3.8, $\approx 87\%$ of candidates are pre-filtered; projected time is 6–10 hours (v5, ARM64, 9 workers).

Open Problem 7.3 (Universality of the witnesses $w_2 = 5$, $w_3 = 7$). In all four certificates, $w_2 = 5$ and $w_3 = 7$ succeeded. Prove (or disprove) that 5 and 7 are valid Pocklington witnesses for every prime of the subfamily.

8 Conclusion

We have given a self-contained treatment of unconditional primality certificates for the 3-smooth subfamily $p = 3m(m + 1) + 1$, $m = 2^a \cdot 3^b - 1$.

Three elementary but effective arithmetic theorems — the mod-6 identity (Theorem 3.1), the cubic reciprocity sieve (Theorem 3.3), and the mod-7 forbidden-class filter (Theorem 3.6) — together eliminate 87% of candidates before any large-number computation. Their proofs rest on standard tools: the Chinese Remainder Theorem, the non-residuacity of -3 modulo primes $q \equiv 2 \pmod{3}$, and the multiplicative orders $\text{ord}_7(2) = 3$, $\text{ord}_7(3) = 6$.

Applied via the PrimeQuest multi-core algorithm on a consumer laptop, this framework produced four unconditional certificates, culminating in a prime of 29 998 decimal digits (286 MR events, 3 h 06 min). A fifth search targeting 50 000 digits is in progress.

Place in the series. As Paper IV of the parametric prime family series [1, 2, 3], this work complements the analytic and distributional results of Papers I–III with a rigorous, computational layer: where the earlier papers characterise the family heuristically or conditionally on Bateman–Horn-type assumptions, the Pocklington subfamily admits unconditional certification. The contrast highlights the value of identifying structurally distinguished sub-classes within an otherwise analytically intractable family.

Acknowledgements

The author thanks the developers of `gmpy2` and GMP for making large-precision arithmetic accessible to independent researchers. Computations were performed on a Dell Inspiron 14 Plus 7441 (Snapdragon X Plus ARM64, 10 Oryon cores, LPDDR5X memory).

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